# Supplementary Material for "Fixed Heavy-Tailed Probabilistic Forecasting under Missing Historical Observations"

[AUTHOR INPUT REQUIRED]

## 1 Full Fixed-Mask Matrix

The following tables retain all fixed-mask deterministic evaluations by dataset. Only historical inputs are masked; targets are unchanged.

Table 1: Full ETTh1 fixed-mask matrix.

| Mask | Rate | MSE | MAE | RMSE |
|---|---|---|---|---|
| Block | 0.10 | 0.388818 | 0.416915 | 0.623552 |
| Block | 0.30 | 0.423837 | 0.451188 | 0.651027 |
| Block | 0.50 | 0.486504 | 0.501242 | 0.697498 |
| Block | 0.70 | 0.591697 | 0.571655 | 0.769219 |
| Channel | 0.10 | 0.479410 | 0.455063 | 0.692394 |
| Channel | 0.30 | 0.583704 | 0.511084 | 0.764006 |
| Channel | 0.50 | 0.798862 | 0.627325 | 0.893791 |
| Channel | 0.70 | 0.900947 | 0.683061 | 0.949182 |
| Mixed | 0.10 | 0.486209 | 0.464128 | 0.697287 |
| Mixed | 0.30 | 0.495055 | 0.473908 | 0.703601 |
| Mixed | 0.50 | 0.553485 | 0.525226 | 0.743966 |
| Mixed | 0.70 | 0.715806 | 0.617339 | 0.846053 |
| Random | 0.10 | 0.378811 | 0.408069 | 0.615476 |
| Random | 0.30 | 0.425394 | 0.450336 | 0.652223 |
| Random | 0.50 | 0.525856 | 0.523115 | 0.725160 |
| Random | 0.70 | 0.688520 | 0.617690 | 0.829771 |

Table 2: Full Weather fixed-mask matrix.

| Mask | Rate | MSE | MAE | RMSE |
|---|---|---|---|---|
| Block | 0.10 | 0.158302 | 0.218956 | 0.397872 |
| Block | 0.30 | 0.180420 | 0.245178 | 0.424759 |
| Block | 0.50 | 0.210474 | 0.277453 | 0.458774 |
| Block | 0.70 | 0.254572 | 0.322297 | 0.504551 |
| Channel | 0.10 | 0.191289 | 0.238851 | 0.437366 |
| Channel | 0.30 | 0.284471 | 0.316355 | 0.533358 |
| Channel | 0.50 | 0.377902 | 0.393866 | 0.614738 |
| Channel | 0.70 | 0.495709 | 0.490618 | 0.704066 |
| Mixed | 0.10 | 0.175815 | 0.230739 | 0.419303 |
| Mixed | 0.30 | 0.224939 | 0.281431 | 0.474278 |
| Mixed | 0.50 | 0.306930 | 0.357264 | 0.554012 |
| Mixed | 0.70 | 0.400193 | 0.440502 | 0.632608 |
| Random | 0.10 | 0.169631 | 0.232543 | 0.411863 |
| Random | 0.30 | 0.240810 | 0.309629 | 0.490724 |
| Random | 0.50 | 0.331313 | 0.391138 | 0.575598 |
| Random | 0.70 | 0.434543 | 0.474367 | 0.659199 |

## 2 TimePerceiver Seed-Level Summaries

The four probability-head seeds are 2021, 2022, 2023, and 2025 on fixed official backbone checkpoints. The panels summarize the dataset-level means and variability; the main manuscript preserves all paired NLL deltas.

## 3 Distribution-Head Ablation Details

The ablation retains Gaussian, Laplace, fixed-df Student-$t$, and learned-df Student-$t$ results. The learned variant is visible even though it is not the final method.

Table 3: Full Electricity fixed-mask matrix.

| Mask | Rate | MSE | MAE | RMSE |
|---|---|---|---|---|
| Block | 0.10 | 0.143247 | 0.243795 | 0.378480 |
| Block | 0.30 | 0.184630 | 0.289155 | 0.429686 |
| Block | 0.50 | 0.270835 | 0.369677 | 0.520418 |
| Block | 0.70 | 0.464169 | 0.520260 | 0.681300 |
| Channel | 0.10 | 0.213972 | 0.281108 | 0.462571 |
| Channel | 0.30 | 0.391038 | 0.403166 | 0.625331 |
| Channel | 0.50 | 0.569116 | 0.525639 | 0.754398 |
| Channel | 0.70 | 0.748916 | 0.649662 | 0.865399 |
| Mixed | 0.10 | 0.165759 | 0.254616 | 0.407135 |
| Mixed | 0.30 | 0.246785 | 0.325159 | 0.496775 |
| Mixed | 0.50 | 0.375182 | 0.437690 | 0.612521 |
| Mixed | 0.70 | 0.588662 | 0.596810 | 0.767243 |
| Random | 0.10 | 0.138429 | 0.242728 | 0.372060 |
| Random | 0.30 | 0.200572 | 0.318891 | 0.447853 |
| Random | 0.50 | 0.325732 | 0.432183 | 0.570729 |
| Random | 0.70 | 0.546080 | 0.584678 | 0.738972 |

## 4 PIT and QQ Diagnostics

These probability integral transform and quantile plots are descriptive diagnostics; they are not formal evidence of joint calibration.

## 5 PatchTST Detailed Results

The second-backbone study uses one backbone seed and one probability-head seed per dataset. ETTh1 and Weather pass the full paired criteria; Electricity preserves NLL and CRPS improvements but exceeds the 1% point-error tolerance.

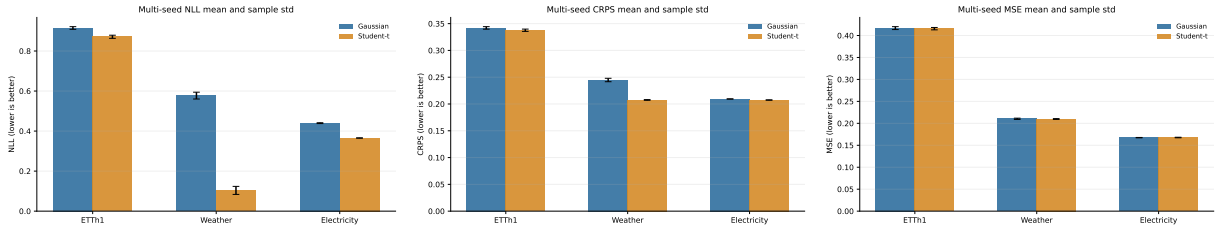



Figure 1: TimePerceiver probability-head seed summaries for NLL, CRPS, and MSE.

Table 4: ETTh1 paired results across additional missing-history conditions. Negative deltas favor fixed Student-$t$.

| Condition | ΔNLL | ΔCRPS | ΔMSE | NLL win | CRPS NI | Point NI |
|---|---|---|---|---|---|---|
| Random 0.1 | -0.078025 | -0.006113 | -0.002231 | Yes | Yes | Yes |
| Random 0.3 | -0.041381 | -0.003864 | -0.001155 | Yes | Yes | Yes |
| Random 0.5 | -0.013720 | 0.000656 | 0.007100 | Yes | No | No |
| Block 0.3 | -0.061086 | -0.007738 | -0.004834 | Yes | Yes | Yes |
| Channel 0.3 | -0.071504 | -0.004849 | 0.001022 | Yes | Yes | Yes |
| Mixed 0.3 | -0.070665 | -0.003286 | 0.001169 | Yes | Yes | Yes |

## 6 Adapter Ablation

The static adapter improves on the official reference in the evaluated ETTh1 condition, while global-statistics FiLM does not improve over matched static capacity.

Table 5: Static and missingness-aware adapter comparison on ETTh1 random-0.3.

| Variant | Trainable | Overhead | Best epoch | Val. MSE | Test MSE | Test MAE |
|---|---|---|---|---|---|---|
| Official | 0 | 0.00% | NA | NA | 0.425394 | 0.450336 |
| Static adapter | 33,600 | 2.96% | 1 | 0.678666 | 0.361377 | 0.390458 |
| Missingness-aware FiLM adapter | 39,136 | 3.44% | 5 | 0.679754 | 0.363156 | 0.391077 |

## 7 Strict Conformal Negative Result

The additive correction uses a temporally separated calibration set and cannot shrink intervals. It widens the already near-nominal native intervals and does not improve absolute coverage error under this protocol.

## 8 Protocol and Reproducibility

The chronological splits, train-only normalization statistics, fixed validation masks, fixed test masks, metric implementations, evaluation seeds, configuration files, and checkpoint hashes are retained in the project evidence registry. Input length is 384, prediction length is 96, and CRPS uses 64 Monte Carlo samples with fixed evaluation seed 52025. Test masks are never used during head training, and test labels are never used for model selection.

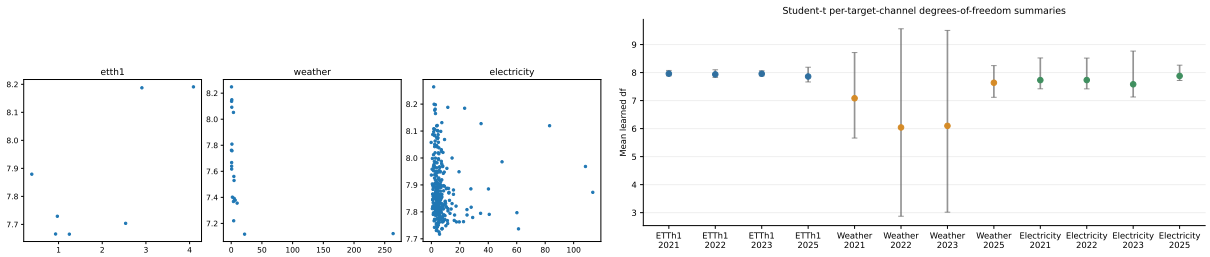



Figure 2: Descriptive channel-tail diagnostics and distribution-head comparison.

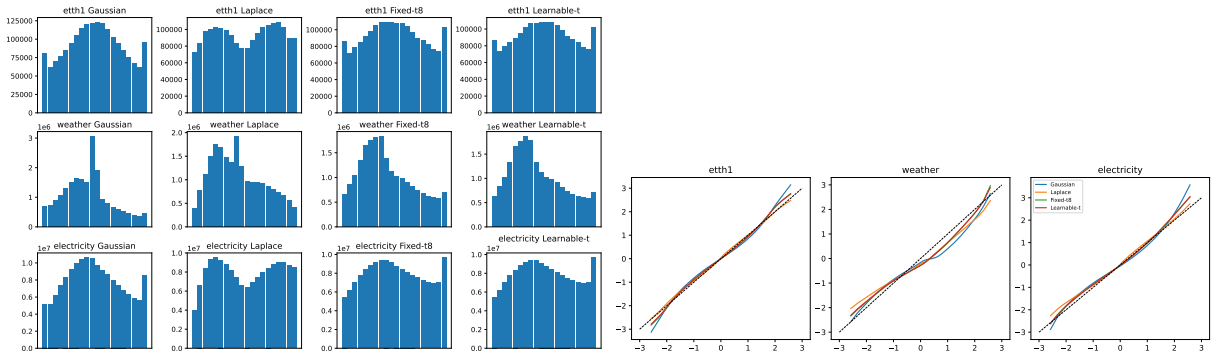



Figure 3: PIT histograms and quantile diagnostics for the distribution ablation.

For traceability, the project stages map to the evidence as follows: the fixed missing-history assets and deterministic evaluation precede the adapter study; probabilistic-head validation and multi-seed fixed-backbone comparisons establish the heavy-tail signal; the distribution ablation refines the final fixed-df specification; the conformal study records a negative result; and PatchTST provides the second-backbone assessment. Historical probability-head runs that did not actually apply the input mask were marked invalid and excluded without reporting their metrics.

## 9 Source and Mask Integrity

Every formal result is linked to a unique experiment record, resolved configuration, checkpoint hash, fixed-mask hash, and metric-protocol hash. The venue adaptation is a document-only transformation of the frozen evidence: it performs no training, inference, metric recomputation, mask generation, or checkpoint modification. Exact repository URLs and anonymous-review release details remain an author action.

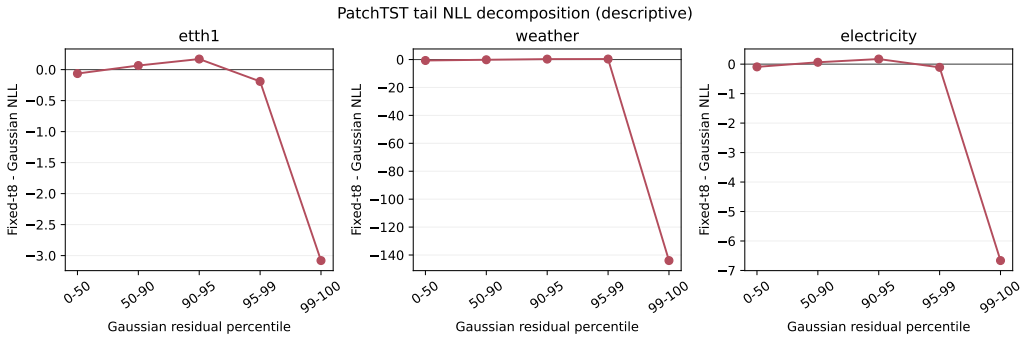



Figure 4: Descriptive PatchTST tail-bin NLL decomposition.

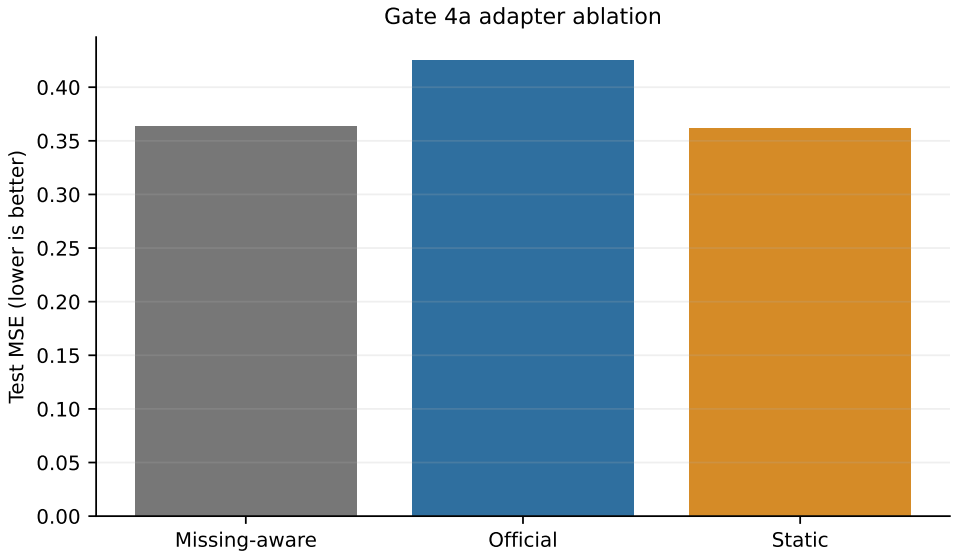



Figure 5: Adapter MSE comparison.

Table 6: Formal additive conformal results.

| Method | Nominal | PICP | MPIW | Coverage error | Interval score |
|---|---|---|---|---|---|
| Mask-severity-conditioned conformal | 0.90 | 0.927136 | 2.262613 | 0.027136 | 2.989775 |
| Marginal conformal | 0.90 | 0.927636 | 2.267989 | 0.027636 | 2.990528 |
| Native | 0.90 | 0.897581 | 1.953189 | 0.002419 | 2.946898 |
| Mask-severity-conditioned conformal | 0.95 | 0.964262 | 2.993500 | 0.014262 | 3.625242 |
| Marginal conformal | 0.95 | 0.964505 | 2.998730 | 0.014505 | 3.625259 |
| Native | 0.95 | 0.937700 | 2.424187 | 0.012300 | 3.596873 |

Table 7: Conformal mechanism audit; counterfactual rows are diagnostic only.

| Head | Nominal | Cal. PICP | Test PICP | Shift | $\hat{q}$ | $\Delta$ coverage error |
|---|---|---|---|---|---|---|
| Gaussian | 0.90 | 0.869114 | 0.904533 | 0.035418 | 0.126842 | 0.023728 |
| Gaussian | 0.95 | 0.912017 | 0.937263 | 0.025246 | 0.279881 | 0.001974 |
| Student-t | 0.90 | 0.861667 | 0.897581 | 0.035913 | 0.157400 | 0.025216 |
| Student-t | 0.95 | 0.913178 | 0.937700 | 0.024522 | 0.287271 | 0.002205 |

Table 8: All retained probability heads and interval procedures.

| Head | Method | Nominal | PICP | MPIW | Coverage error | Interval score |
|---|---|---|---|---|---|---|
| Gaussian | Mask-severity-conditioned conformal | 0.90 | 0.927797 | 2.289391 | 0.027797 | 2.979145 |
| Gaussian | Marginal conformal | 0.90 | 0.928261 | 2.294196 | 0.028261 | 2.979930 |
| Gaussian | Native | 0.90 | 0.904533 | 2.040512 | 0.004533 | 2.936425 |
| Gaussian | Mask-severity-conditioned conformal | 0.95 | 0.964530 | 2.986506 | 0.014530 | 3.598434 |
| Gaussian | Marginal conformal | 0.95 | 0.964711 | 2.991181 | 0.014711 | 3.598406 |
| Gaussian | Native | 0.95 | 0.937263 | 2.431420 | 0.012737 | 3.571486 |
| Student-t | Mask-severity-conditioned conformal | 0.90 | 0.927136 | 2.262613 | 0.027136 | 2.989775 |
| Student-t | Marginal conformal | 0.90 | 0.927636 | 2.267989 | 0.027636 | 2.990528 |
| Student-t | Native | 0.90 | 0.897581 | 1.953189 | 0.002419 | 2.946898 |
| Student-t | Mask-severity-conditioned conformal | 0.95 | 0.964262 | 2.993500 | 0.014262 | 3.625242 |
| Student-t | Marginal conformal | 0.95 | 0.964505 | 2.998730 | 0.014505 | 3.625259 |
| Student-t | Native | 0.95 | 0.937700 | 2.424187 | 0.012300 | 3.596873 |

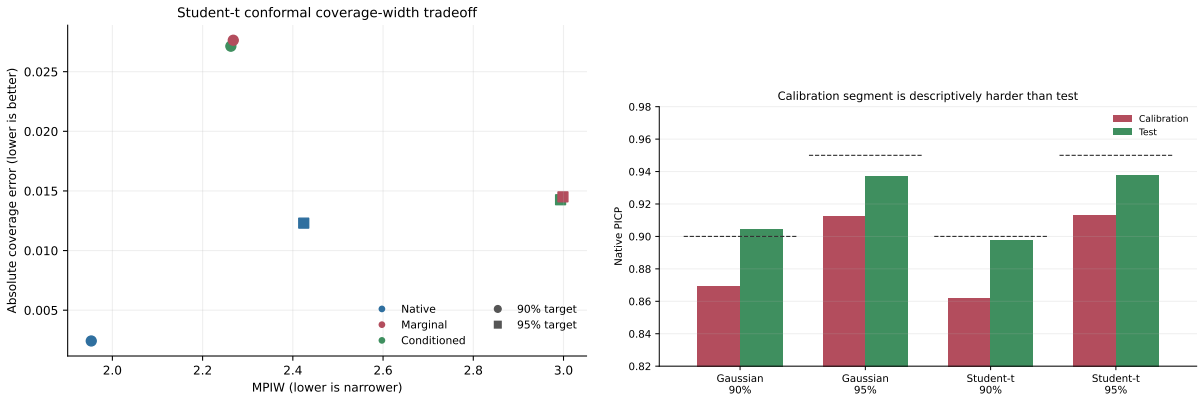



Figure 6: Interval-width tradeoff and calibration-to-test coverage shift.

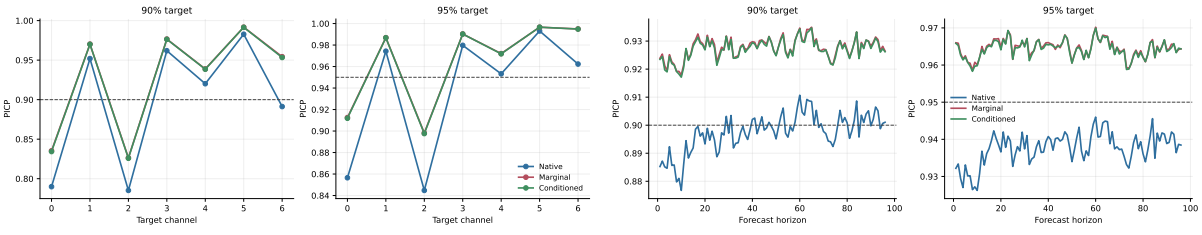



Figure 7: Channel- and horizon-level marginal coverage diagnostics.